What is a charge?

3 basic measurements

L - length
M - mass
t - time
new! → Charge → not the same as electron

Coulomb's Law

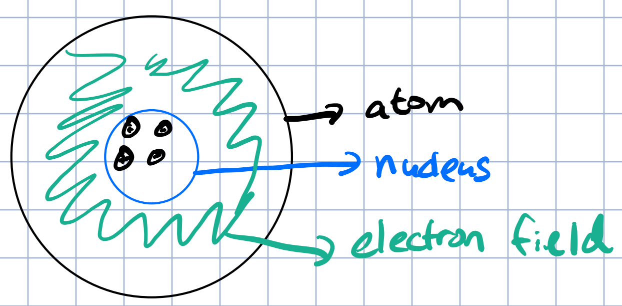
$\oplus \xrightarrow{F_{12}} r - \oplus$
$q_1 \qquad\qquad q_2$

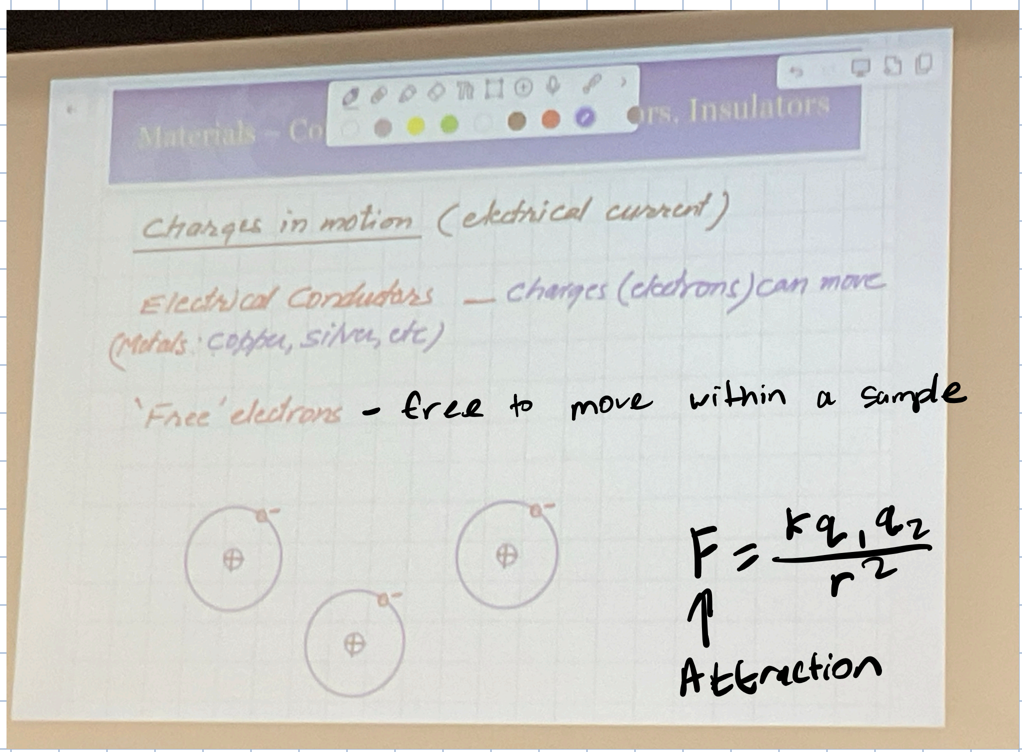
$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \rightarrow \text{magnitude}$$

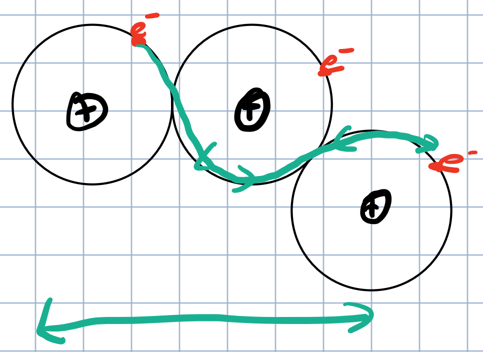
Force of 1 on 2

$$\frac{1}{4\pi\epsilon_0} = k = 9 \cdot 10^9 \, N \cdot m^2/C$$

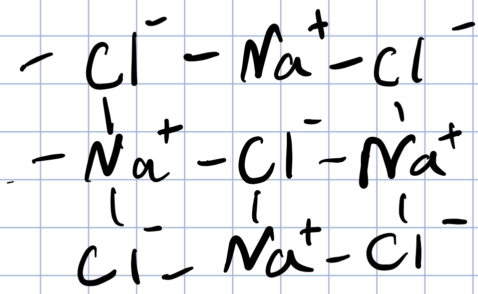
Like charges repel unlike charges attract

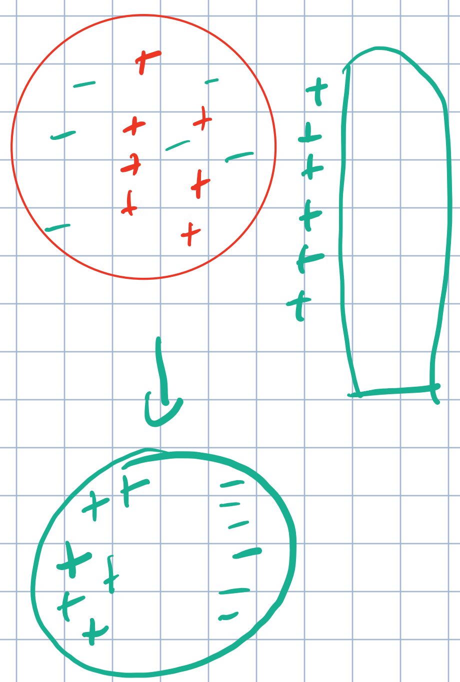






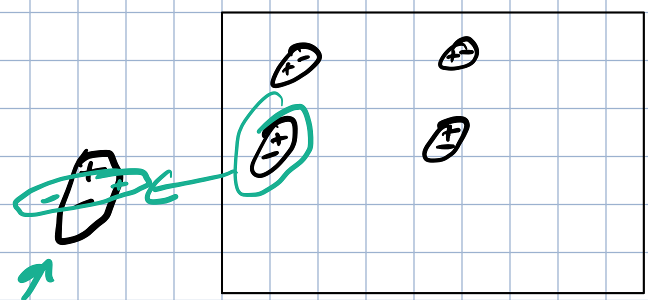
electrons move between a metal





Induced.

for the can/rod



rotate

electrical insulator

paper cap

Coulomb's Law

$\underset{q_1}{\oplus}$ — r — $\underset{q_2}{\oplus}$

$$F = \frac{k q_1 q_2}{r^2}$$

Same charge repel

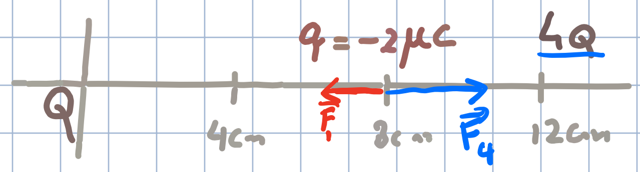
$F_{12} \leftarrow \overset{1}{\oplus} - r - \overset{2}{\oplus} \rightarrow F_{12}$

Quiz

$$\boxed{\vec{F}_{12} = \vec{F}_{21}}$$

ex1 PROB. 21 47

$\vec{F} = 126N\hat{i}$ $\quad$ $Q = ?$



$$\vec{F}_{net} = \vec{F}_4 + \vec{F}_1$$

$$F_{net} = \frac{kq(4Q)}{(4\cdot10^{-2})^2} - \frac{kqQ}{(8\cdot10^{-2}m)^2}$$

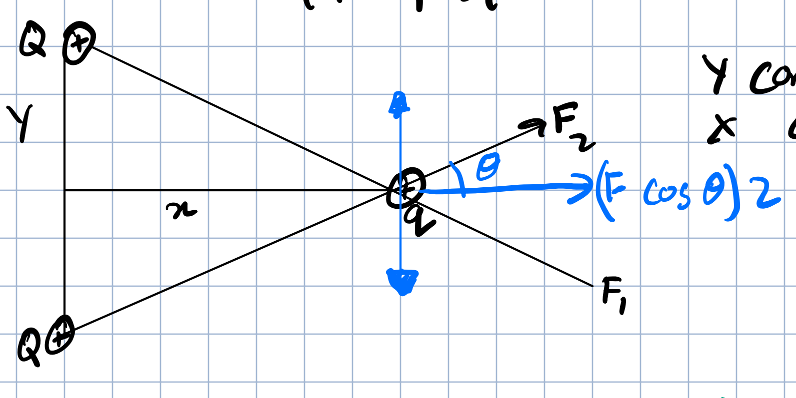
they repel

$$= \frac{kqQ}{(4\cdot10^{-2}m)^2}\left(\frac{4}{4}\right) = \frac{kqQ(3.75)}{(4\cdot10^{-2})^2} = 126N$$

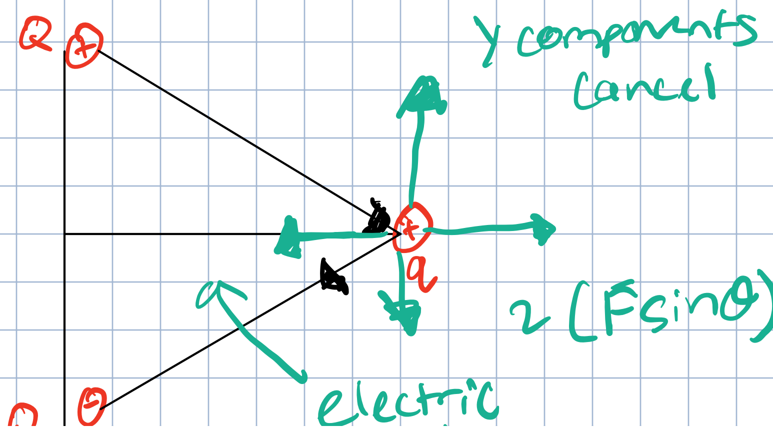
$$Q = \frac{126(4\cdot10^{-2})^2}{(9\cdot10^{9})(2\cdot10^{-6})\cdot3.75} \approx \boxed{3}$$

ex2

$$|F_1| = |F_2| = F$$



Y components cancel
X components add



dipolic

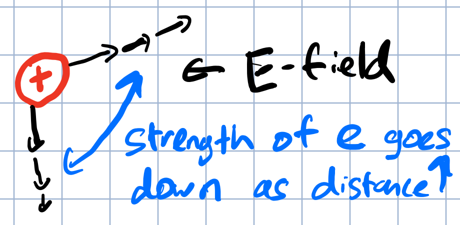
Day 2

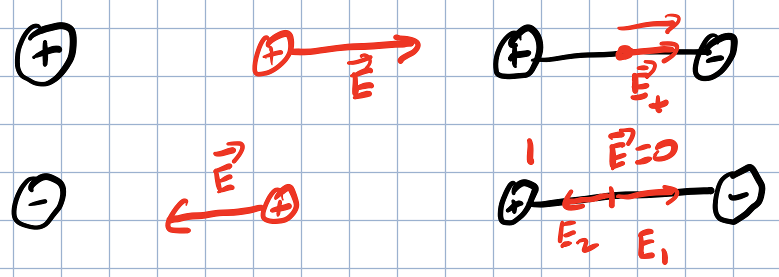
Newton's law of gravitation

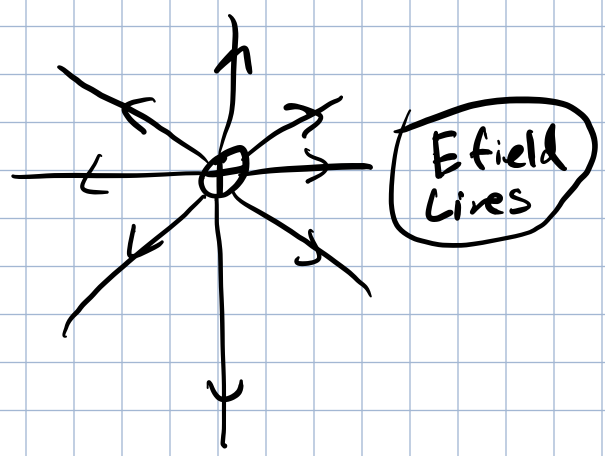
$$F = \frac{k q_1 q_2}{R^2} = \left(\frac{k q_1}{R^2}\right) q_2$$

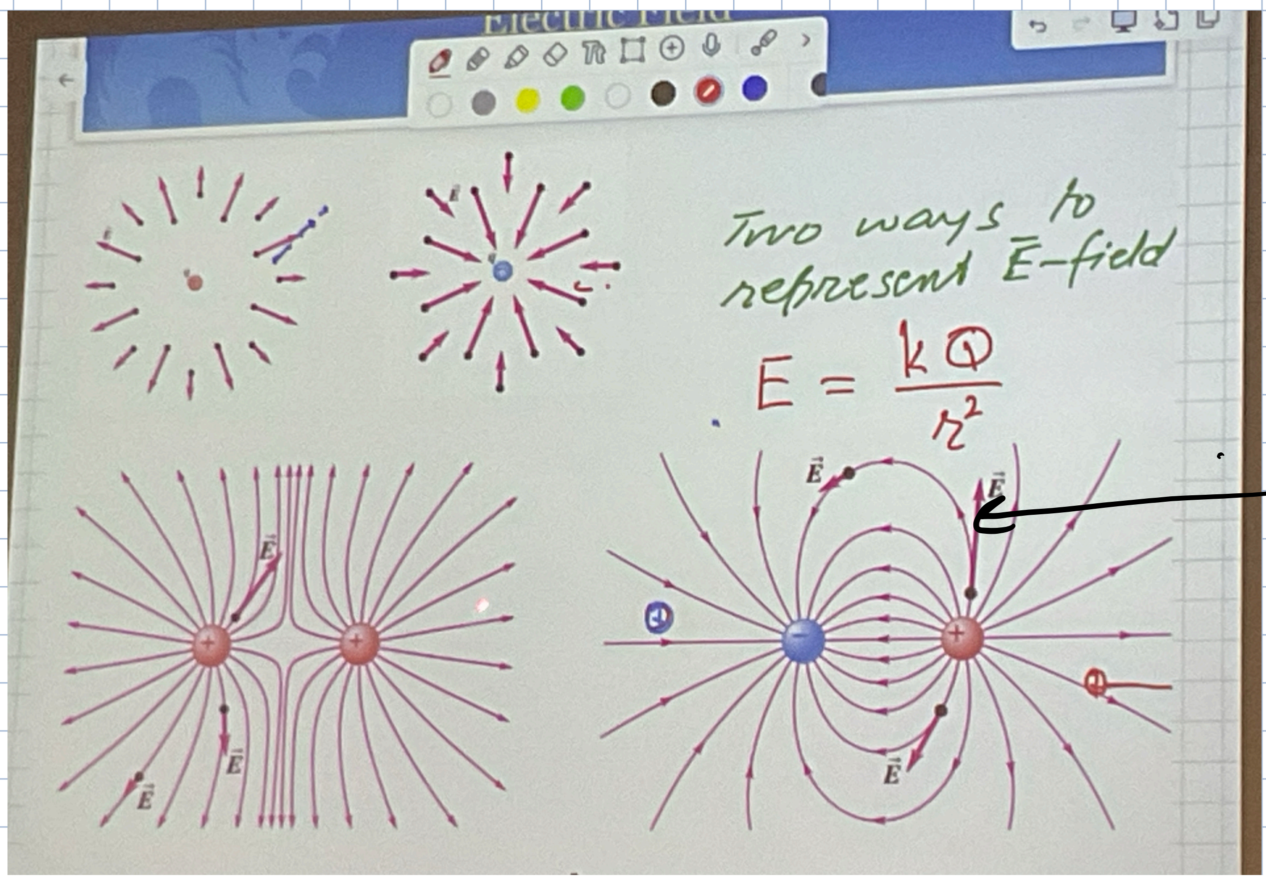
field → region of influence

O_{q_1} $O_{q_2} \leftarrow \frac{k q_1}{R_2}$



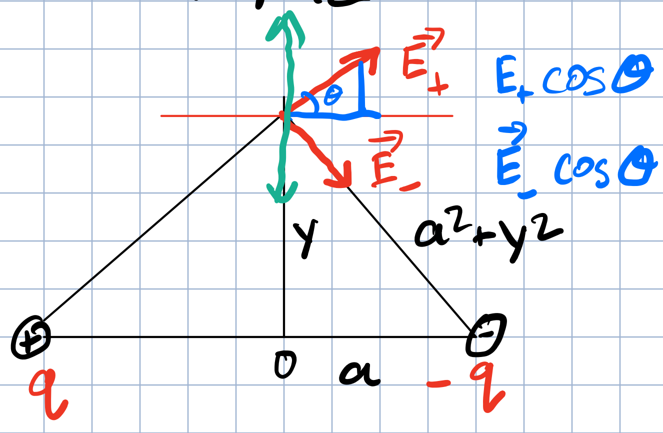






tangent line

ex 1. Dipole



$|\vec{E}_+| = |\vec{E}_-| = E$

$E_{net} = E_+ \cos\theta + E_- \cos\theta$

$= 2E \cos\theta$

y components cancel

$E_{net} = 2E \cos\theta = \frac{2kq}{a^2+y^2} \cdot \frac{a}{(a^2+y^2)^{1/2}}$

$= \frac{2kqa}{(a^2+y^2)^{3/2}}$

$p = (2a)q$

Dipole moment

$\oplus \xleftarrow{\vec{p}} \ominus$

$\vec{E}_{net} = \frac{2kqa}{(a^2+y^2)^{3/2}}$ @ $y=0$

$= \frac{2kq}{a^2}$

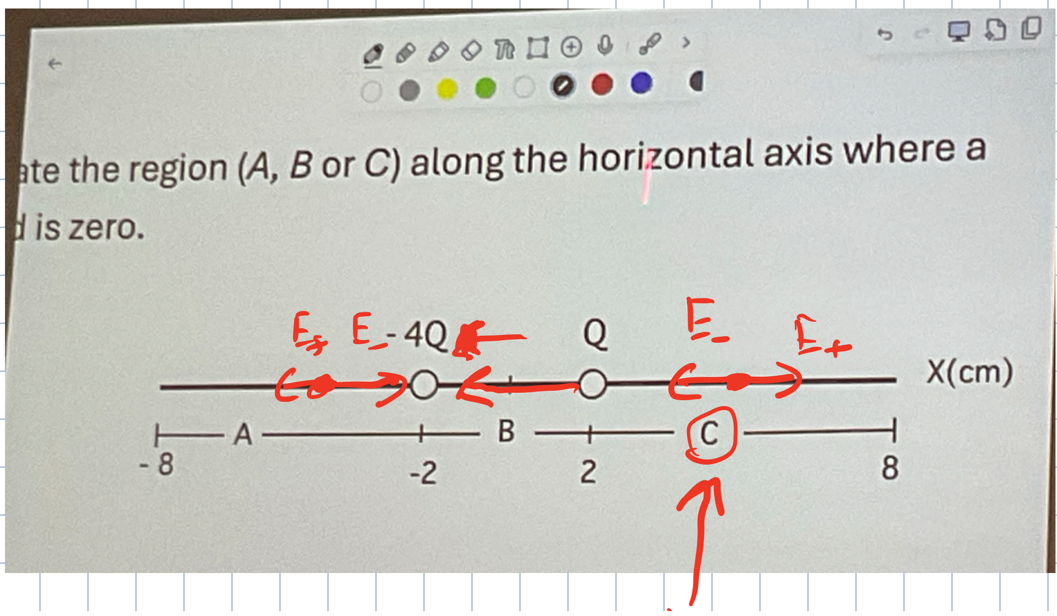
If

$y >>> a$

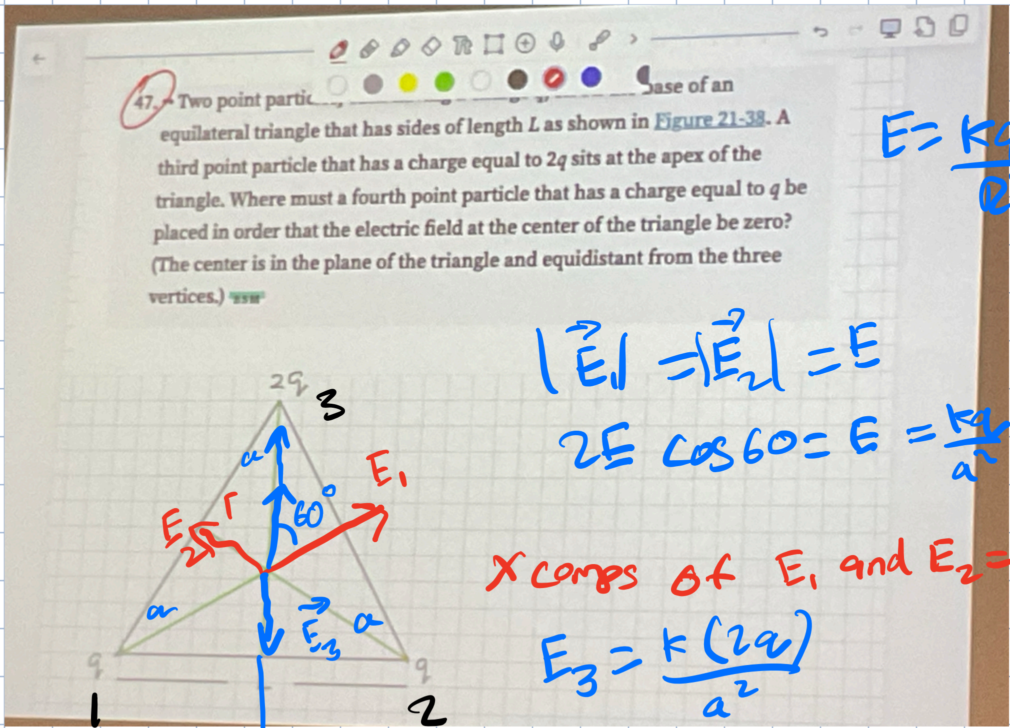
much bigger

$E_{net} = \frac{2kqa}{y^3} \to 0$

E field goes to 0 if the charges go far away



They will be 0 here


47. Two point partic[illegible]base of an equilateral triangle that has sides of length L as shown in Figure 21-38. A third point particle that has a charge equal to $2q$ sits at the apex of the triangle. Where must a fourth point particle that has a charge equal to q be placed in order that the electric field at the center of the triangle be zero? (The center is in the plane of the triangle and equidistant from the three vertices.)

$E = \frac{kq}{a^2}$

$|\vec{E}_1| = |\vec{E}_2| = E$

$2E\cos 60 = E = \frac{kq}{a^2}$

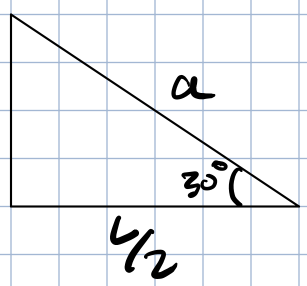
x comps of E_1 and $E_2 = 0$

$E_3 = \frac{k(2q)}{a^2}$

q

$\vec{E}_{net} = -\frac{kq}{a^2}\hat{j}$

Should be placed $a = L/\sqrt{3}$ below P



$a\cos 30 = L/2$ $\quad a = L/\sqrt{3}$